Lab #4 - Longitudinal Modes
ECE 482/582, Fall 2003

Goals:

1. To measure the longitudinal mode structure of a HeNe laser using a Fabry-Perot Interferometer.
2. To measure the true linewidth of the HeNe laser.
3. To gain hands on understanding of how a scanning Fabry-Perot Interferometer works.
4. To get an experimental and intuitive feel for the following: free spectral range (FSR), finesse, and gain bandwidth.

Reading:

Kuhn, *Laser Engineering*, Chapter 1: section 1.9 and Chapter 3

Other Resources:

Moore, John, *Building Scientific Apparatus*, Section 4.7.4 – 4.7.5

Java applet - http://www.physics.uq.edu.au/people/mcintyre/applets/optics/fabry.html

Preparatory Questions:

1. Calculate the free spectral range, finesse, and the Q for a Fabry-Perot Interferometer consisting of 2 flat mirrors of reflectivity 99.8% and an air spacing of 3cm. The laser is a HeNe of wavelength 632.8nm.

2. If one of the mirrors from question 1 is now scanned over a distance of 2um, how many different Fabry-Perot resonances will be covered?

3. If the HeNe laser has 2 lasing longitudinal modes separated by 500MHz in frequency, one with higher intensity than the other, sketch the scanning Fabry-Perot output expected.

4. If the HeNe laser cavity has mirrors separated by 30cm, how will the FP output change?

5. If the spacing of the two FP mirrors is now increased to 4.5 cm, how will the FP output change? Sketch the output expected.

Laboratory Experiment:

There is one scanning Fabry-Perot Interferometer (FPI) available for your use. One mirror is fixed and the other moves on a piezoelectric transducer cylinder driven by a voltage ramp. The movements are very small (as you should expect if you answered the preparatory questions) and you cannot see them. The whole cylinder mount can be moved manually. This device can also be used to explore the behavior of Fabry-Perot laser cavities as a function of mirror spacing, type and scan length.

Procedure:

1. Set the scanning FPI mirror spacing to 6cm.
2. Measure the mode structure of the HeNe laser and sketch your results in your notebook. Determine the finesse of the cavity at this length.
3. Observe the mode structure as a function of laser temperature (observe slow changes as the laser tube temperature drifts with the room temperature). Document your findings.
4. Observe the mode structure as a function of the laser polarization using a sheet of linear polarizer material). Document your findings.
5. Change the length of the FP cavity by a small amount (<1cm) using the lead screw and measure the change in mode spacing.
6. Change the length of the FP cavity using the lead screw to 4.5cm and measure the mode spacing. Determine the finesse of the cavity at this length.
7. Change the length of the FP cavity using the lead screw to 3.0cm and measure the mode spacing. Determine the finesse of the cavity at this length.
8. Replace one of the FP flat mirrors with a curved mirror and repeat the measurements made for steps 2-7. You may need to take extra care in aligning the laser with the FP. Make sure to sketch observations in your notebook.

Questions to Answer:

- Was the finesse of the FPI different for different mirror spacing, or the same?
- Using your measures results for the FSR of the HeNe laser cavity, calculate the length of the HeNe laser cavity inside the black metal tube.
- How were the modes affected by temperature? Are they polarized? Describe the polarization states.
- Calculate the FSR of the Fabry-Perot Interferometer with 6cm and 4.5cm mirror spacing. This will be the scale factor of the oscilloscope trace of signal vs time.
- Assuming the gain bandwidth of a typical HeNe laser to be 1.5GHz, how many longitudinal modes might you expect to lase?
- The flat FP mirrors have a reflectivity of 99.7%. What is the calculated finesse assuming no losses? What is the best finesse you measured and how did you determine it?
- What causes the measured finesse to be lower than the calculated finesse?

Conclusion:

You should now have an understanding of the theory and use of Fabry-Perot cavities. You should also be familiar with a FPI and know how to use it to measure the mode structure of a laser.